01 FINDINGS

What did Develop change?

Both arms passed their case-specific acceptance checks in all five completed historical pairs. None showed a production-behavior advantage for Develop.

+9.6%

release-to-final time

+51.2%

total context tokens

Develop relative to control, using summed measurements across five cases. Time includes review and waiting. Context includes cached input.

Two of eight selected cases are complete

01-02 Requests and Cobra. 03-08 Six cases still lack valid pairs.

Which five cases contributed to the totals?

Requests and Cobra are in the final slate. Starlette, Flask and Hugo are supplemental historical pairs. Each supplies one paired observation. All five are Python or Go projects.

How far can the findings travel?

The target changed from 20 cases to eight after some outcomes were known. Roux chose the design and interpreted the results. They warrant independent reassessment.

02 METHOD

One task, two isolated arms

Compare Codex with and without Develop under the same review process. The merged human PR supplies a third reference for behavior and patch quality.

CONTROL	DEVELOP	EVALUATOR
Clean base and repository guidance. Develop is inaccessible.	Same clean base and guidance, plus the frozen Develop package.	Separate gold checkout, hidden acceptance tests and arm mapping.

1. Freeze and admit each case

Verify the gold merge and its first parent, requirement and toolchain hashes. Run base and gold checks offline. Keep the worker prompt free of PR identifiers and solution hints.

2. Match execution and test the boundaries

Use CLI 0.159.2, gpt-6.1-sol at medium effort and Develop 0.9.0. Run arms serially with equal resources and time limits. Each gets one synthetic base commit. Deny command egress, sibling, evaluator and auth paths for workers, reviewers and descendants.

3. Measure the full workflow

Each arm in the five completed pairs used one implementation root, two planning reviewers and two final reviewers. Start the clock before instruction loading. Reconcile final cumulative usage once per native thread. Retain waits, retries, partials and missing values.

4. Grade anonymous patches

Check acceptance and new regressions. Hide arm identities and timing from graders. Compare behavior, tests, security and maintenance against gold. Use targeted reproductions for disagreements and retain uncertainty when evidence is inconclusive.

Limits include possible style leakage or memorization, prior gold exposure for acceptance authors, and retrospective tool-event auditing.

03 MEASURED RESOURCES

What did each pair use?

The same five historical pairs appear in both charts. Each bar starts at zero. Time and context have separate scales.

FIVE-PAIR SUM	SECONDS	UNCACHED IN	CACHED IN	OUTPUT	CONTEXT
Control	1,879.6	667,538	3,708,288	45,506	4,421,332
Develop	2,059.9	887,476	5,729,536	66,291	6,683,303

Output includes reasoning. Totals exclude coordinator, grader and human gold usage, which are unavailable. The ledger does not establish billed cost.

04 QUALITY EVIDENCE

How did the patches compare?

Acceptance is measured per case. Qualitative preferences below come from the historical blind assessments. Different test sets make pooled test counts misleading.

Requests

SELECTED / 7 OF 7

Both handled file-like proxies and passed 243 socket-free native checks, with 13 skipped. The assessors found mixed maintenance and test tradeoffs. Full HTTP coverage was incomplete.

Cobra

SELECTED / 13 LEAF SUBCASES

Both satisfied the validator requirement and passed the full Go suite. Gold retained more contextual diagnostics. The two agents added complementary regression tests.

Starlette

SUPPLEMENTAL / 12 OF 12

Both agents preserved a repeated-slash root-path boundary that gold broke. Their production fixes matched. The improvement over gold was reproduced after the trials.

Flask

SUPPLEMENTAL / 17 OF 17

Both preserved explicit host and port overrides with an invalid configured port. Gold raised an error in this post-hoc check. Assessors modestly preferred Develop's regression tests.

Hugo

SUPPLEMENTAL / 6 API CASES AND 1 REAL TEMPLATE

Runtime behavior matched across both arms. Each improved template regression coverage over gold. Assessors favored Develop's additional pipeline and conversion boundary tests.

Only the tested paths support these comparisons.

05 FINAL SLATE

Where do the eight cases stand?

The first five projects are external. The last three are user-owned and have prior-context risk. Each case label links to its merged gold PR.

CASE / GOLD PR	FIRST / LIMIT	LAST VERIFIED STATE
Requests #7505 Complete	Control 40 min / arm	Both arms passed all seven acceptance checks.
Cobra #2397 Complete	Develop 20 min / arm	Both arms passed 13 leaf subcases and the full Go suite.
Tokio #8686 Prepared	Control 20 min / arm	No arms. Worker baseline 26/26. The new API was absent on base.
fmt #4993 Interrupted	Develop 20 min / arm	Develop partial passed 13/13. Final review hit a service limit. Control unstarted.
Svelte #19076 Prepared	Develop 75 min / arm	No arms. Independent checks passed 8/8 on both references. Gold-derived checks went from 0/2 to 2/2.
Personal site #413 Not admitted	Control 75 min / arm	Offline install and build passed. Five browser projects still need runtime admission.
law-config #20 Interrupted	Develop 75 min / arm	Develop stopped before edits. Control unstarted. Acceptance was 2/7 on base and 7/7 on gold.
Runner-space #50 Limited preparation	Control 75 min / arm	No arms. Behavioral checks went from 3/9 to 9/9. Six native permission failures affected both references.

Six selected pairs remain. fmt and law-config each retain one fresh complete-pair retry. The separate Vue attempt exhausted its retry.

06 DECISIONS

What should Develop keep or test?

The record supports specific test practices and a new cost experiment. It cannot establish broader quality gains or savings from shorter reviews.

Why the benchmark stopped

The original run paused for handoff after repeated service limits, and its scheduled restart was canceled. The ledger retains 20 attempted arms and 10 human references. Five attempted arms have only lower-bound usage.

The continuation recovered all 16 selected base/gold archives and the pinned Linux CLI. Four representative Tokio role profiles passed 14 synthetic checks each, both directly and in descendants. This covered one configuration.

Device sign-in was unavailable. The pinned app server opened a thread without an authenticated account. Desktop builds differed from the frozen CLI. Execution stopped at the user's request before a scored turn.

Recommendations

01 **Keep requirement-linked tests.**
Preserve boundary cases like Flask's overrides and Hugo's pipeline inputs. Measure the behavior each test protects.

02 **Measure review costs by phase.**
Record planning, implementation and review usage before testing smaller packets in a separately versioned comparison.

03 **Prove the fixture distinguishes the change.**
Freeze behavior criteria before release. Identify gold-derived tests and keep ambiguous pilots visible.

04 **Qualify one complete matched pilot.**
Verify same-account inference, isolation, instruction loading and per-request accounting. Retain failures and unknowns.